

EVANSVILLE, IN, USA

WMO: 724320

Lat: 38.044N

Lon: 87.521W

Elev: 122

StdP: 99.87

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93817

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.1	-10.1	-18.7	0.7	-11.6	-15.9	1.0	-9.1	11.0	4.2	9.6	4.3	2.8	320	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	34.3	24.3	33.0	24.1	31.9	23.7	26.1	32.0	25.5	31.1	24.9	30.0	3.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	19.8	29.4	23.9	19.0	28.6	23.3	18.3	28.0	81.4	32.0	78.7	31.1	75.9	30.0	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.1	7.2	DB	-16.9	36.2	3.5	1.7	-19.5	37.4	-21.5	38.4	-23.5	39.3	-26.0	40.6	(4)
(5)			WB	-17.4	27.4	3.4	0.7	-19.9	27.9	-21.9	28.3	-23.8	28.6	-26.3	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	0.8	3.0	8.0	14.0	19.3	24.0	25.6	25.1	21.3	14.8	8.0	3.1		
	DBStd	10.04	6.54	6.16	5.86	5.05	4.30	3.16	2.63	2.67	3.87	4.89	5.29	5.79		
	HDD10.0	968	294	208	109	19	1	0	0	0	12	100	224			
	HDD18.3	2418	544	430	322	144	42	3	0	1	17	131	312	473		
	CDD10.0	2418	8	12	47	141	290	421	483	467	338	161	39	10		
	CDD18.3	827	0	0	3	16	73	174	225	209	105	22	1	0		
	CDH23.3	8001	0	1	21	134	626	1696	2272	2042	993	211	7	0		
	CDH26.7	3118	0	0	1	16	174	676	963	857	381	49	1	0		
(14) Wind	WSAvg	3.0	3.6	3.6	3.6	3.6	3.1	2.7	2.4	2.2	2.2	2.7	3.1	3.2		
(15) Precipitation	PrecAvg	1132	74	81	116	116	124	96	105	76	81	76	98	96		
	PrecMax	1775	232	257	326	299	343	192	234	216	222	209	215	209		
	PrecMin	708	13	15	34	29	23	4	5	6	4	0	13	14		
	PrecStd	212	44	52	60	67	71	47	54	50	55	46	51	46		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	21.3	25.8	28.5	31.9	34.9	36.5	36.2	34.6	30.3	24.2	20.0		
		MCWB	15.6	15.7	17.4	19.1	22.5	24.5	25.7	24.8	22.0	21.2	17.5	16.4		
	2%	DB	16.3	18.4	23.0	26.8	30.1	33.2	34.0	34.1	32.6	28.0	21.3	16.9		
		MCWB	14.2	14.0	16.1	18.4	21.4	23.9	25.2	24.4	22.3	20.0	15.9	14.1		
	5%	DB	13.9	15.7	20.4	24.9	28.7	31.9	32.7	32.5	30.8	25.8	19.0	14.3		
		MCWB	11.9	12.0	14.5	17.4	20.9	23.5	24.8	24.0	21.9	18.4	14.6	11.8		
	10%	DB	10.9	13.0	17.8	22.9	27.1	30.6	31.4	31.1	28.8	23.4	16.7	12.0		
		MCWB	8.3	9.4	12.7	16.4	20.3	22.9	24.1	23.4	20.8	17.4	12.9	9.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.3	17.0	18.5	21.3	24.1	26.1	27.2	26.9	24.7	22.9	19.0	17.0		
		MCDB	18.0	20.2	24.4	26.4	29.1	32.4	32.8	33.1	30.6	28.2	22.0	19.5		
	2%	WB	14.6	14.9	16.9	19.9	23.0	25.4	26.4	25.9	23.9	21.5	17.2	14.7		
		MCDB	16.3	17.5	21.4	24.8	28.1	31.3	32.2	31.7	29.5	25.9	20.3	16.4		
	5%	WB	12.2	12.7	15.4	18.6	22.0	24.6	25.7	25.2	23.0	19.9	15.3	12.2		
		MCDB	13.8	15.2	19.4	23.0	27.0	30.2	31.2	30.6	28.3	23.8	18.3	13.9		
	10%	WB	8.5	9.5	13.5	17.4	21.1	23.9	25.1	24.5	22.2	18.4	13.3	9.7		
		MCDB	10.5	12.7	17.3	21.6	25.5	29.0	30.2	29.4	26.7	22.5	16.2	11.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	9.4	10.5	11.6	10.8	10.5	10.3	10.9	12.2	11.9	10.3	8.4		
		MCDBR	10.9	13.3	13.6	13.3	12.2	11.8	11.5	12.3	13.9	14.1	13.0	10.7		
	5% WB	MCWBR	9.4	10.0	8.2	7.4	5.6	4.6	4.5	4.6	5.2	6.8	8.8	8.7		
		MCWBR	10.1	12.0	12.0	11.2	10.5	10.4	10.1	10.5	11.2	11.6	11.2	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.327	0.367	0.408	0.440	0.451	0.474	0.481	0.417	0.362	0.329	0.310		
		taud	2.455	2.399	2.352	2.242	2.203	2.195	2.141	2.111	2.281	2.429	2.469	2.470		
	Ebn at Noon		866	901	893	872	846	834	810	795	829	849	842	845		
		Edh at Noon	82	99	115	136	144	145	152	153	122	95	81	76		
(44) All-Sky Solar Radiation	RadAvg		1.97	2.68	3.74	4.95	5.56	6.36	6.11	5.62	4.66	3.44	2.35	1.70		
	RadStd		0.21	0.31	0.34	0.28	0.48	0.40	0.39	0.30	0.33	0.36	0.25	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	+78		

Nomenclature: See separate page